

# Ethan Wu

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## EDUCATION

### Queen's University

*Bachelor of Applied Science, Mechatronics and Robotics Engineering*  
– GPA: 3.89/4.3 | Dean's Scholar

Kingston, ON

Sept. 2024 – April 2028

## EXPERIENCE

### Teaching Assistant (APSC 101)

Aug. 2026 – Present

*Stephen J. R. Smith Faculty of Engineering and Applied Science*

Kingston, ON

- Instructing 40+ 1st year engineering students on the iterative design process, Fusion modeling, and physical prototyping for a kidney dialysis system.
- Facilitating studio sessions by guiding student teams of 4-6 through Arduino debugging and systematic circuit integration.

### Automation Team Manager

Oct. 2025 – Present

*Queen's Engineering Society*

Kingston, ON

- Promoted to lead a student team automating digital infrastructures, workflows, and data management for EngSoc administration and services, including Campus Outfitters, Clark Hall Pub, Orientation Week, etc.
- Build custom flows and applications using Power Automate and other web tools to eliminate manual tracking for service directors and executive leads.
- Manage team hiring, project onboarding, and task delegation to scale/support upcoming software projects.

### Embedded Systems Member

Sept. 2025 – June 2026

*Queen's AutoDrive Team*

Kingston, ON

- Developed a C++ memory monitoring module for an embedded watchdog to detect system crashes.
- Earned 2nd Place in the Mobility Innovation Challenge for the SAE AutoDrive Competition for proposing an AI driver training curriculum designed to accelerate adoption of AVs.

### QEC Challenges Coordinator & QSC Finance Coordinator

June 2026 – Present

*Queen's Engineering Challenge/Queen's Space Conference*

Kingston, ON

- Designed, drafted, and budgeted the QEC Senior Design Challenge rubric and delegate package for QEC 2026.
- Building an automated physical arena to simulate pop-up fires and a fire suppression turret prototype to test QEC Senior Design challenge feasibility.
- Oversee budget allocations, expense tracking, and financial forecasting for QSC operations.

## PROJECTS

### TriPAD - Construction Site Sensor | C++, Arduino, Sensor Fusion

Apr. 2026 – Aug. 2026

- Built and assembled a standalone, automated environmental monitoring unit named the TriPAD using an ESP32 to process real-time data from dust (PMS5003), noise (MAX9814), and vibration (MPU6050) sensors.
- Designed a custom 3D-printed Autodesk Fusion enclosure for tripod deployment.
- Awarded 3rd place at the General's Lab Innovation Project hosted by the OGCA and American Global for the TriPAD.

### Mars Sample Return (MSR) Rover | ROS2, C++, Python, R&D

Jan. 2026 – Apr. 2026

- Developed a ROS2-based autonomous rover with a bidirectional Raspberry Pi-to-Arduino bridge.
- Implemented closed-loop PI control with sensor fusion (IMU/encoders) for precise motion correction.
- Integrated RPLIDAR A1 for real-time environment mapping and Nav2 stack localization.

### Autonomous Transforming Robot | Autodesk Fusion, Kinematics

Jan. 2026 – Present

- Designing a 30+ DOF kinematic assembly (Optimus Prime) with interference checks and motion analysis.
- Optimizing mechanical linkages for 3D printing and refining tolerances for geared actuators.
- Developing transition algorithms to maintain center-of-mass stability during physical reconfiguration.

## TECHNICAL SKILLS

**Software & Controls:** C/C++, Python, ROS2, Nav2, SLAM, PID/PI Control, Sensor Fusion, Kinematics, SQL, Linux

**Hardware & Embedded:** ESP32, Raspberry Pi, Arduino, CAN bus, UART, I2C, PMS5003, MAX9814, MPU6050, LiDAR, Encoders

**CAD & Tools:** Autodesk Fusion, SOLIDWORKS, LTSpice, Quartus, MATLAB, Power Automate, Git, LaTeX; Cantonese, Mandarin, French

**Hands-On:** Soldering, Mechanical Assembly Welding, Machining, Woodworking, 3D Printing